

Can EARTHQUAKES, TSUNAMIS, And VOLCANO ERUPTIONS Be Predicted WELL IN ADVANCE Now?

An experimental setup running in India was able to detect ionospheric changes that were associated with the volcanic eruption that occurred on 15th January this year in Tonga. The precursor nature could be observed as early as four weeks starting 17th December last year!



PRASANNA WAICHAL

holds a master's degree in electronics and has more than 25 years of work experience in industrial R&D for design and development of specialised and strategic electronics systems and academic teaching and research at post-graduate levels, both in India and abroad. His passions include radio reception and studying solar activities and their effect on ionosphere on earth. He is also a licensed amateur radio operator with call-sign VU3QOI. He claims to have discovered a method to predict earthquakes using very-low-frequency (VLF) multipath-multi-frequency propagation technique

In January 2022, there was a sudden underwater volcanic eruption in the Kingdom of Tonga, a tiny island country in the South Pacific region. The event was captured by satellites and there have been reports of atmospheric, acoustic, and pressure waves observed following this event. This article reports for the first time from India about the ionospheric effects observed using very low frequency (VLF) radio wave propagation associated with this volcanic eruption.

The basics

The earth's atmosphere is divided into two parts, the lower atmosphere that is close to the earth's surface and the outer side that opens into the space and can be called the upper atmosphere. It is this upper atmosphere that gets ionised from the Sun's x-rays and extreme ultra-violet radiations. Positive ions and negative charges (electrons) are generated as part

of this ionisation process. This ionised layer is called the ionosphere.

Depending upon the density and height of the atmosphere, there exists a gradient of number of electrons generated as a function of height. This leads further to various layers having different electron densities. Starting from the bottom to upward direction, the D layer is at approximately 60km-90km height. Next, the E layer is at around 90km-150km height, while F1 and F2 layers are at 150km-250km and 250-350km height approximately. During night, since ionisation stops, the recombination starts and the D layer vanishes while the F1 and F2 merge into a single F-layer.

Depending upon the frequency of operation in the LF/MF and HF range (30kHz to 30MHz), some layers absorb the radio signals while others reflect the signals. If a frequency is still higher, it simply passes through

the ionosphere to the outer space. The D layer absorbs or attenuates most of the radio signals in HF band but the E, F1, and F2 layers reflect some of the frequencies. This varies depending upon time of the day, dawn and dusk conditions, etc.

The VLF signals

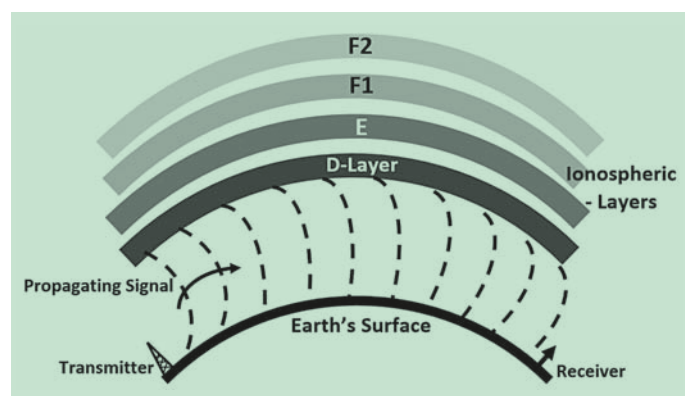


Fig. 1: The various layers in the ionosphere and the VLF radio propagation using the Earth-Ionosphere-Waveguide Mode